

PS1 - IO

Tate Mason

```
# Packages
library(tidyverse)
```

```
-- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
v dplyr      1.1.4      v readr      2.1.5
v forcats    1.0.0      v stringr    1.5.1
v ggplot2    3.5.2      v tibble     3.3.0
v lubridate  1.9.4      v tidyr      1.3.1
v purrr      1.1.0
-- Conflicts ----- tidyverse_conflicts() --
x dplyr::filter() masks stats::filter()
x dplyr::lag()     masks stats::lag()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become
```

```
library(lubridate)

# Data
iri <- read_tsv("~/SchoolWork/Y2S1/IO/PS1/IRI.csv", show_col_types = FALSE)
names(iri) <- gsub("^t", "", names(iri)) # Clean names
iri <- iri %>%
  mutate(
    market_id = interaction(store_id, week_id, drop = TRUE),
    date_id = interaction(year, month, drop = TRUE),
    date = as.Date(paste(year, month, "01", sep = "-"))
  )
```

1.1

```

# Distinct counts per market
brands_per_market <- iri %>%
  distinct(market_id, brand) %>%
  count(market_id, name = "n_brands")

mfrs_per_market <- iri %>%
  distinct(market_id, parent) %>%
  count(market_id, name = "n_mfrs")

# Total market sales per market
total_sales_market <- iri %>%
  group_by(market_id) %>%
  summarise(total_sales = sum(quantity, na.rm = TRUE), .groups = "drop")

# Brand-level sales (market x brand)
brand_sales_market <- iri %>%
  group_by(market_id, brand) %>%
  summarise(brand_sales = sum(quantity, na.rm = TRUE), .groups = "drop")

# Per-market price & characteristics (means within market)
market_chars <- iri %>%
  group_by(market_id) %>%
  summarise(
    price_mean      = mean(price, na.rm = TRUE),
    fiber_mean      = mean(fiber, na.rm = TRUE),
    sugars_mean     = mean(sugars, na.rm = TRUE),
    flavored_mean   = mean(as.numeric(flavored), na.rm = TRUE),
    fortified_mean  = mean(as.numeric(fortified), na.rm = TRUE),
    .groups = "drop"
  )

# Helper to summarise numeric vectors
summarise_vec <- function(x) {
  tibble(
    mean  = mean(x, na.rm = TRUE),
    sd    = sd(x, na.rm = TRUE),
    median = median(x, na.rm = TRUE),
    max   = max(x, na.rm = TRUE)
  )
}

# Tables for 1.1

```

```

tbl_counts <- bind_cols(
  variable = c("n_brands", "n_mfrs"),
  bind_rows(
    summarise_vec(brands_per_market$n_brands),
    summarise_vec(mfrs_per_market$n_mfrs)
  )
)

tbl_total_sales <- bind_cols(
  variable = "total_sales (per market)",
  summarise_vec(total_sales_market$total_sales)
)

tbl_brand_sales <- bind_cols(
  variable = "brand_sales (per market-brand)",
  summarise_vec(brand_sales_market$brand_sales)
)

tbl_price_chars <- tribble(
  ~variable, ~mean, ~sd, ~median, ~max,
  "price (market mean)",
    summarise_vec(market_chars$price_mean)$mean,
    summarise_vec(market_chars$price_mean)$sd,
    summarise_vec(market_chars$price_mean)$median,
    summarise_vec(market_chars$price_mean)$max,
  "fiber (market mean)",
    summarise_vec(market_chars$fiber_mean)$mean,
    summarise_vec(market_chars$fiber_mean)$sd,
    summarise_vec(market_chars$fiber_mean)$median,
    summarise_vec(market_chars$fiber_mean)$max,
  "sugars (market mean)",
    summarise_vec(market_chars$sugars_mean)$mean,
    summarise_vec(market_chars$sugars_mean)$sd,
    summarise_vec(market_chars$sugars_mean)$median,
    summarise_vec(market_chars$sugars_mean)$max,
  "flavored share (market mean)",
    summarise_vec(market_chars$flavored_mean)$mean,
    summarise_vec(market_chars$flavored_mean)$sd,
    summarise_vec(market_chars$flavored_mean)$median,
    summarise_vec(market_chars$flavored_mean)$max,
  "fortified share (market mean)",
    summarise_vec(market_chars$fortified_mean)$mean,

```

```

    summarise_vec(market_chars$fortified_mean)$sd,
    summarise_vec(market_chars$fortified_mean)$median,
    summarise_vec(market_chars$fortified_mean)$max
  )

summary_11 <- bind_rows(
  tbl_counts,
  tbl_total_sales,
  tbl_brand_sales,
  tbl_price_chars
)

summary_11

```

```

# A tibble: 9 x 5
  variable                mean      sd  median    max
  <chr>                <dbl>   <dbl>   <dbl>   <dbl>
1 n_brands              34.9    4.82    37      39
2 n_mfrs                 3.95    0.259    4       4
3 total_sales (per market) 15555. 12358. 12557. 84567.
4 brand_sales (per market-brand) 446.    726.   238. 26470.
5 price (market mean)      0.235   0.0265  0.233   0.325
6 fiber (market mean)      1.30    0.0874  1.31    1.67
7 sugars (market mean)     8.31    0.205   8.33    10.2
8 flavored share (market mean) 0.640   0.0265  0.641   0.857
9 fortified share (market mean) 0.485   0.0341  0.487   0.667

```

```

# If you want a file:
# write_csv(summary_11, "summary_11_descriptives.csv")

```

1.2

```

ts_simple <- iri %>%
  group_by(week_id) %>%
  summarise(
    avg_price = mean(price, na.rm = TRUE),
    avg_qty   = mean(quantity, na.rm = TRUE),
    .groups = "drop"
  ) %>%

```

```

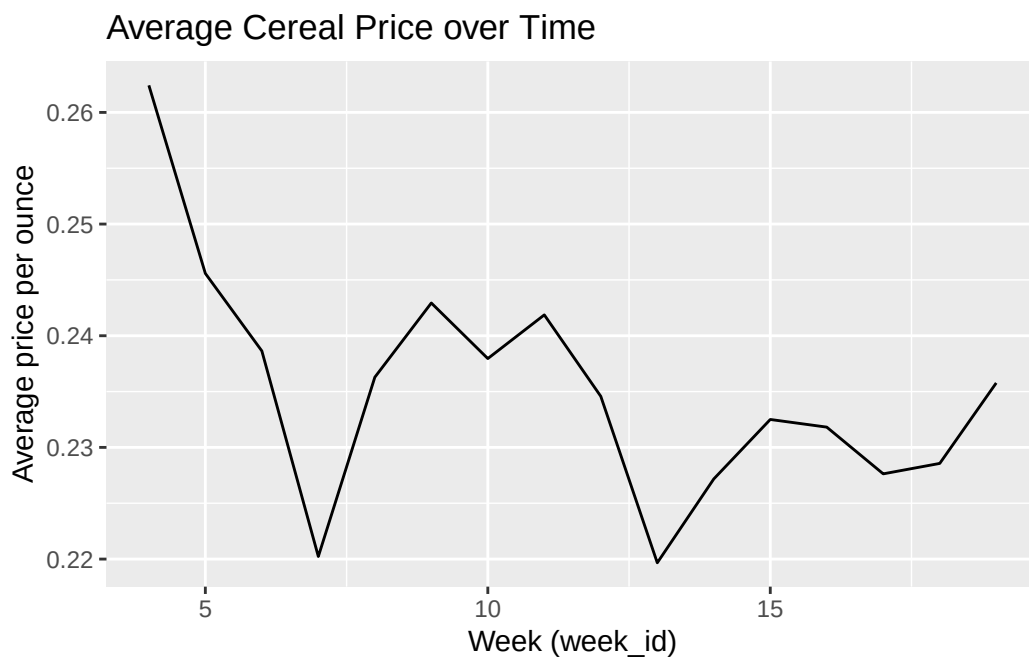
arrange(week_id)

p_price <- ggplot(ts_simple, aes(x = week_id, y = avg_price)) +
  geom_line() +
  labs(title = "Average Cereal Price over Time",
       x = "Week (week_id)", y = "Average price per ounce")

p_qty <- ggplot(ts_simple, aes(x = week_id, y = avg_qty)) +
  geom_line() +
  labs(title = "Average Cereal Sales (Quantity) over Time",
       x = "Week (week_id)", y = "Average quantity (lbs)")

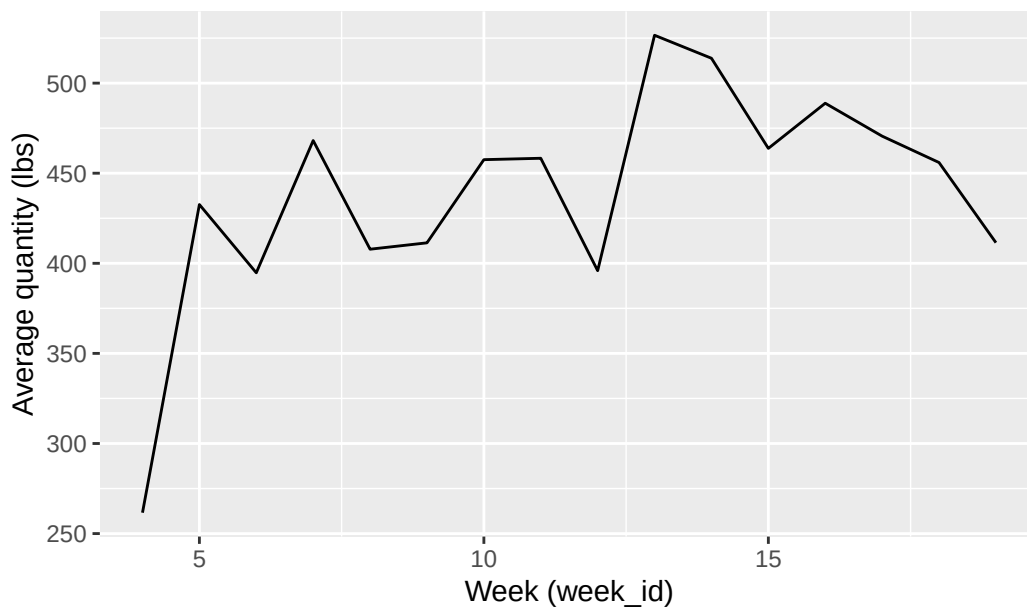
p_price

```



```
p_qty
```

Average Cereal Sales (Quantity) over Time

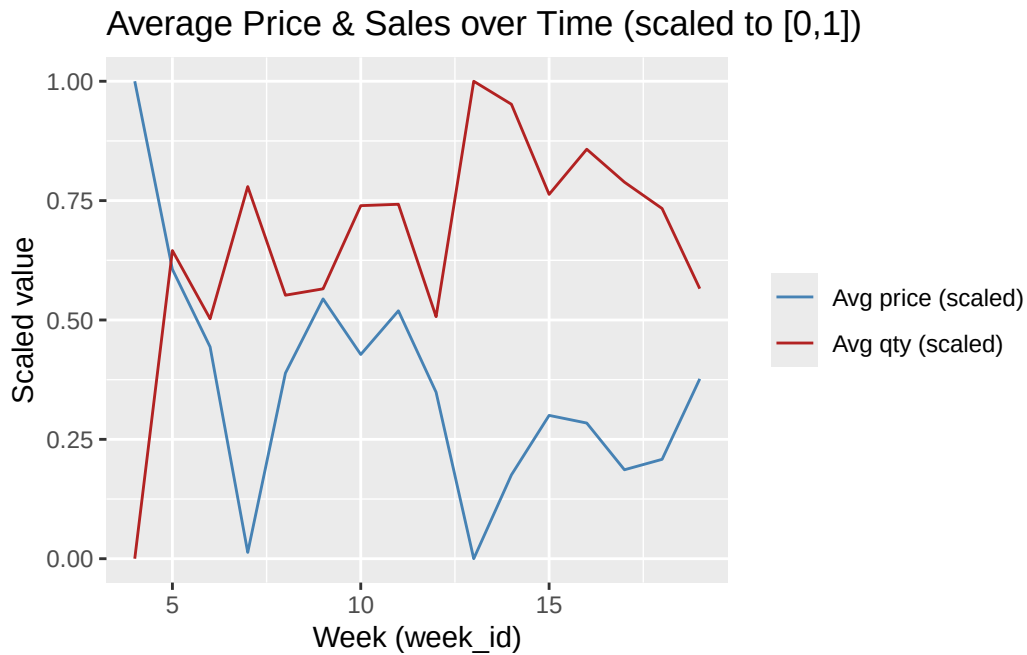


```
# Save figures
ggsave("avg_price.png", p_price, width = 10, height = 6, dpi = 300)
ggsave("avg_sales.png", p_qty, width = 10, height = 6, dpi = 300)

# Dual-axis style (scaled)
ts_long_scaled <- ts_simple %>%
  mutate(
    price_scaled = (avg_price - min(avg_price, na.rm = TRUE)) /
      (max(avg_price, na.rm = TRUE) - min(avg_price, na.rm = TRUE)),
    qty_scaled = (avg_qty - min(avg_qty, na.rm = TRUE)) /
      (max(avg_qty, na.rm = TRUE) - min(avg_qty, na.rm = TRUE))
  ) %>%
  select(week_id, price_scaled, qty_scaled) %>%
  pivot_longer(-week_id, names_to = "series", values_to = "value")

p_dual <- ggplot(ts_long_scaled, aes(x = week_id, y = value, color = series)) +
  geom_line() +
  scale_color_manual(values = c("price_scaled" = "steelblue", "qty_scaled" = "firebrick"),
    labels = c("Avg price (scaled)", "Avg qty (scaled)")) +
  labs(title = "Average Price & Sales over Time (scaled to [0,1])",
    x = "Week (week_id)", y = "Scaled value", color = NULL)

p_dual
```



```
ggsave("avg_price_sales_dual.png", p_dual, width = 10, height = 6, dpi = 300)
```

1.3

```
# Brand HHI per market
brand_shares <- iri %>%
  group_by(market_id) %>%
  mutate(total_mkt_sales = sum(quantity, na.rm = TRUE)) %>%
  ungroup() %>%
  group_by(market_id, brand) %>%
  summarise(brand_sales = sum(quantity, na.rm = TRUE),
            total_mkt_sales = first(total_mkt_sales),
            .groups = "drop") %>%
  mutate(share_brand = if_else(total_mkt_sales > 0, brand_sales / total_mkt_sales, 0))

hhi_brand_market <- brand_shares %>%
  group_by(market_id) %>%
  summarise(HHI_brand = sum(share_brand^2, na.rm = TRUE), .groups = "drop")

hhi_brand_week <- iri %>%
  distinct(market_id, week_id) %>%
```

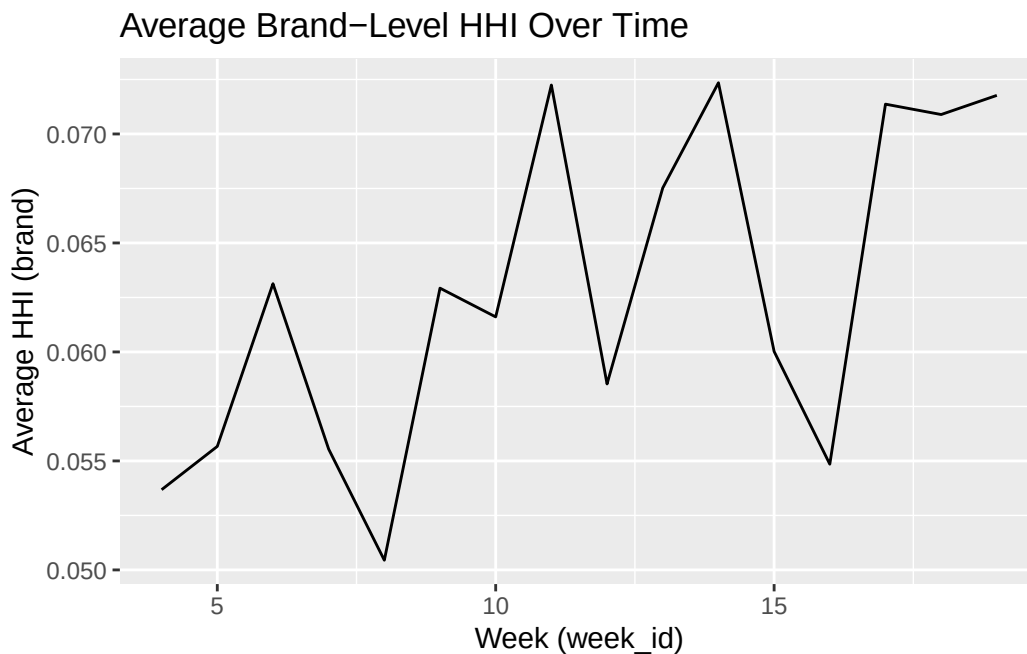
```

inner_join(hhi_brand_market, by = "market_id") %>%
group_by(week_id) %>%
summarise(avg_HHI_brand = mean(HHI_brand, na.rm = TRUE), .groups = "drop") %>%
arrange(week_id)

p_hhi_brand <- ggplot(hhi_brand_week, aes(x = week_id, y = avg_HHI_brand)) +
  geom_line() +
  labs(title = "Average Brand-Level HHI Over Time",
       x = "Week (week_id)", y = "Average HHI (brand)")

p_hhi_brand

```



```

ggsave("HHI_brand.png", p_hhi_brand, width = 10, height = 6, dpi = 300)

# Manufacturer (parent) HHI per market
parent_shares <- iri %>%
  group_by(market_id) %>%
  mutate(total_mkt_sales = sum(quantity, na.rm = TRUE)) %>%
  ungroup() %>%
  group_by(market_id, parent) %>%
  summarise(parent_sales = sum(quantity, na.rm = TRUE),
            total_mkt_sales = first(total_mkt_sales),
            .groups = "drop") %>%

```



```

mutate(share_parent = if_else(total_mkt_sales > 0, parent_sales / total_mkt_sales, 0))

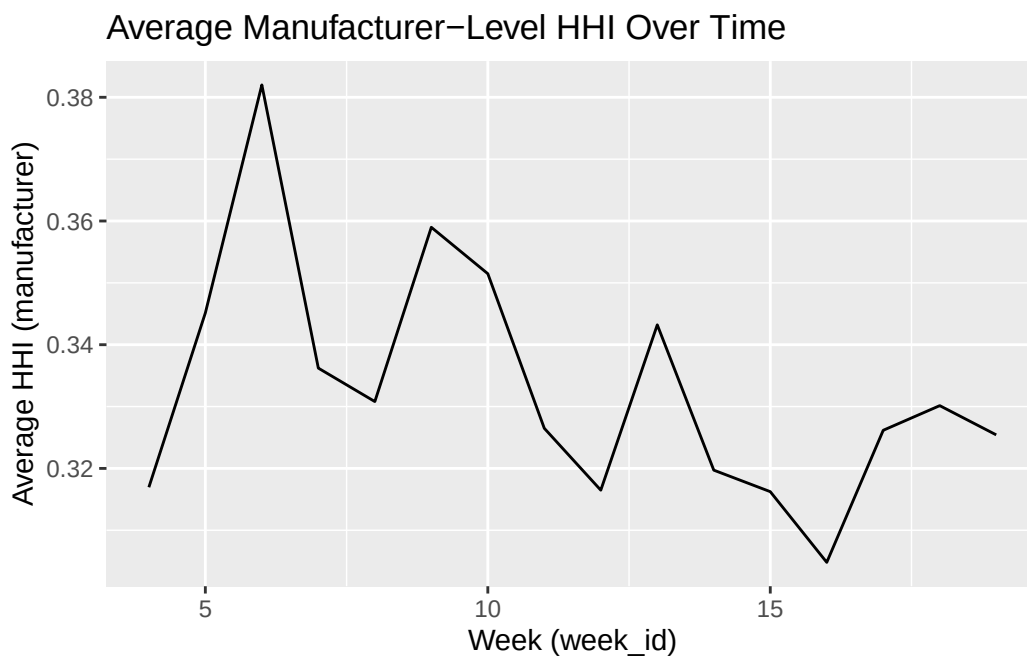
hhi_parent_market <- parent_shares %>%
  group_by(market_id) %>%
  summarise(HHI_parent = sum(share_parent^2, na.rm = TRUE), .groups = "drop")

hhi_parent_week <- iri %>%
  distinct(market_id, week_id) %>%
  inner_join(hhi_parent_market, by = "market_id") %>%
  group_by(week_id) %>%
  summarise(avg_HHI_parent = mean(HHI_parent, na.rm = TRUE), .groups = "drop") %>%
  arrange(week_id)

p_hhi_parent <- ggplot(hhi_parent_week, aes(x = week_id, y = avg_HHI_parent)) +
  geom_line() +
  labs(title = "Average Manufacturer-Level HHI Over Time",
       x = "Week (week_id)", y = "Average HHI (manufacturer)")

p_hhi_parent

```



```

ggsave("HHI_parent.png", p_hhi_parent, width = 10, height = 6, dpi = 300)

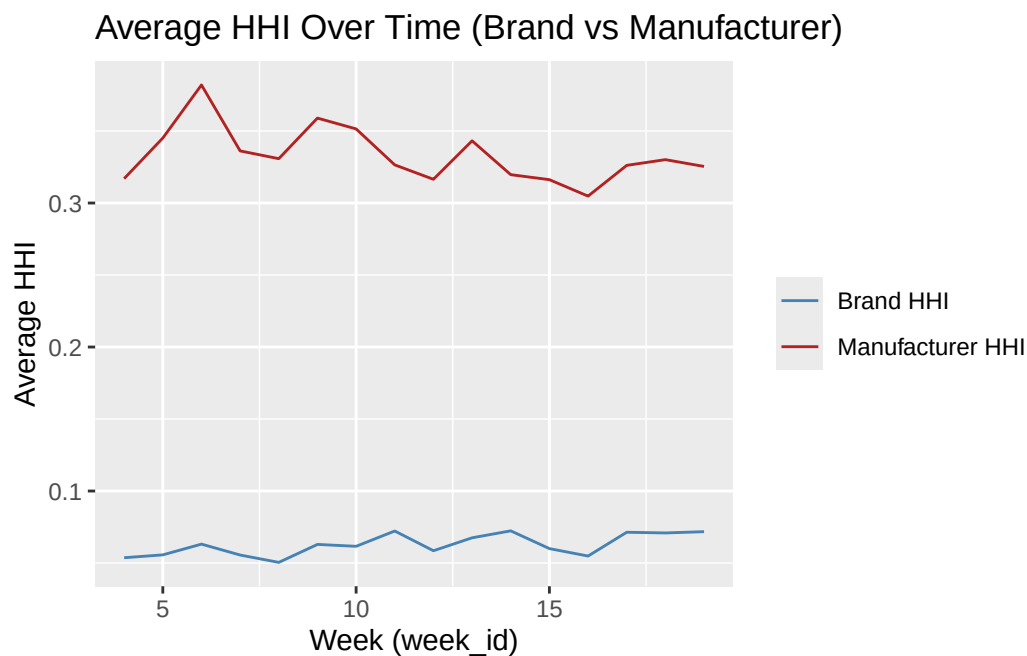
# Combined comparison

```

```
hhi_both <- hhi_brand_week %>% inner_join(hhi_parent_week, by = "week_id")

p_hhi_both <- ggplot(hhi_both, aes(x = week_id)) +
  geom_line(aes(y = avg_HHI_brand, color = "Brand HHI")) +
  geom_line(aes(y = avg_HHI_parent, color = "Manufacturer HHI")) +
  scale_color_manual(values = c("Brand HHI" = "steelblue", "Manufacturer HHI" = "firebrick")) +
  labs(title = "Average HHI Over Time (Brand vs Manufacturer)",
       x = "Week (week_id)", y = "Average HHI", color = NULL)

p_hhi_both
```



```
ggsave("HHI_brand_vs_parent.png", p_hhi_both, width = 10, height = 6, dpi = 300)
```

Question 2:

```
# Prepare data for regression
reg_data <- iri %>%
  group_by(market_id) %>%
  mutate(
    market_fe = as.factor(state),
```

```

manufacturer_fe = as.factor(parent),
time_fe = as.factor(week_id),
brand_fe = as.factor(brand),
p_jmt = price,
total_sales = sum(quantity, na.rm = TRUE),
sj = quantity / M,
s0 = 1 - (total_sales / M),
u = log(sj) - log(s0)
) %>%
ungroup()

```

Our estimation model is thus:

$$u_{jmt} = \alpha p_{jmt} + \beta \mathbb{X}_{jmt} + \xi_{jmt}$$

where $\mathbb{X}_{jmt}$ includes the product characteristics fiber, sugars, flavored, and fortified, as well as fixed effects for market, manufacturer, and time. and u_{jmt} is the log-odds of product j's market share in market m at time t.

```

# Regression model
reg_model <- lm(u ~ p_jmt + fiber + sugars + flavored + fortified +
               market_fe + manufacturer_fe + time_fe + brand_fe,
               data = reg_data)
summary(reg_model)

```

Call:

```
lm(formula = u ~ p_jmt + fiber + sugars + flavored + fortified +
    market_fe + manufacturer_fe + time_fe + brand_fe, data = reg_data)
```

Residuals:

Min	1Q	Median	3Q	Max
-5.0934	-0.4252	0.0600	0.4958	3.7968

Coefficients: (7 not defined because of singularities)

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-0.4054702	0.0638422	-6.351	2.15e-10
p_jmt	-6.6389158	0.0802566	-82.721	< 2e-16
fiber	-0.9442654	0.0164697	-57.333	< 2e-16
sugars	-0.4002088	0.0054006	-74.105	< 2e-16
flavored	0.7597779	0.0191629	39.648	< 2e-16

fortified	0.7032186	0.0181086	38.833	< 2e-16
market_feMassachusetts	-0.1051903	0.0084919	-12.387	< 2e-16
market_feTexas	-0.1519766	0.0103205	-14.726	< 2e-16
market_feIllinois	-0.0375639	0.0100223	-3.748	0.000178
market_feIndiana	-0.4650618	0.0264689	-17.570	< 2e-16
manufacturer_feKELLOGGS	3.3521849	0.0617187	54.314	< 2e-16
manufacturer_fePEPSICO INC	0.7512321	0.0230711	32.562	< 2e-16
manufacturer_feRALCORP HOLDINGS	1.4637079	0.0296752	49.324	< 2e-16
time_fe5	-0.0001525	0.0247749	-0.006	0.995090
time_fe6	-0.1727962	0.0231846	-7.453	9.23e-14
time_fe7	-0.0941736	0.0261641	-3.599	0.000319
time_fe8	0.0617927	0.0262584	2.353	0.018613
time_fe9	-0.0859918	0.0238900	-3.599	0.000319
time_fe10	-0.0888300	0.0230981	-3.846	0.000120
time_fe11	-0.1818979	0.0238172	-7.637	2.25e-14
time_fe12	-0.2368606	0.0236620	-10.010	< 2e-16
time_fe13	-0.2548369	0.0245400	-10.385	< 2e-16
time_fe14	-0.2969521	0.0232892	-12.751	< 2e-16
time_fe15	-0.2451704	0.0237494	-10.323	< 2e-16
time_fe16	-0.3108623	0.0257226	-12.085	< 2e-16
time_fe17	-0.3356226	0.0238612	-14.066	< 2e-16
time_fe18	-0.4433923	0.0233000	-19.030	< 2e-16
time_fe19	-0.3943541	0.0231974	-17.000	< 2e-16
brand_feGENERAL MILLS HONEY NUT CHEER	3.2169077	0.0345498	93.109	< 2e-16
brand_feGENERAL MILLS KIX	-1.4719961	0.0482058	-30.536	< 2e-16
brand_feGENERAL MILLS LUCKY CHARMS	1.0693120	0.0318262	33.598	< 2e-16
brand_feGENERAL MILLS MULTI GRAIN CHE	1.3007579	0.0295959	43.951	< 2e-16
brand_feGENERAL MILLS REESES PUFFS	0.7201245	0.0354311	20.325	< 2e-16
brand_feGENERAL MILLS TOTAL	1.7422502	0.0512161	34.018	< 2e-16
brand_feKELLOGGS APPLE JACKS	-0.6250416	0.0376848	-16.586	< 2e-16
brand_feKELLOGGS CORN FLAKES	-4.1907697	0.0915611	-45.770	< 2e-16
brand_feKELLOGGS CORN POPS	-0.7650356	0.0514965	-14.856	< 2e-16
brand_feKELLOGGS FROOT LOOPS	-1.4175250	0.0403043	-35.171	< 2e-16
brand_feKELLOGGS FROSTED FLAKES	-0.8320666	0.0513888	-16.192	< 2e-16
brand_feKELLOGGS FROSTED MINI WHEATS	-0.5542465	0.0349417	-15.862	< 2e-16
brand_feKELLOGGS RAISIN BRAN	1.1400337	0.0329477	34.601	< 2e-16
brand_feKELLOGGS RICE KRISPIES	-5.9856226	0.0943644	-63.431	< 2e-16
brand_feKELLOGGS SPECIAL K	-4.3832269	0.0807080	-54.310	< 2e-16
brand_feKELLOGGS SPECIAL K RED BERRIE	-3.4856503	0.0875502	-39.813	< 2e-16
brand_fePOST COCOA PEBBLES	-0.9864852	0.0403148	-24.470	< 2e-16
brand_fePOST FRUITY PEBBLES	-0.5247379	0.0401997	-13.053	< 2e-16
brand_fePOST GRAPE NUTS	-0.5977234	0.0277653	-21.528	< 2e-16
brand_fePOST HONEY BUNCHES OF OATS	0.1962830	0.0393273	4.991	6.02e-07

brand_feQUAKER CAP N CRUNCH	0.6339845	0.0303206	20.909	< 2e-16
brand_feQUAKER OATMEAL SQUARES	1.3757840	0.0292445	47.044	< 2e-16
brand_feGENERAL MILLS COCOA PUFFS	0.7582075	0.0354349	21.397	< 2e-16
brand_feGENERAL MILLS GOLDEN GRAHAMS	-1.2770623	0.0268852	-47.501	< 2e-16
brand_feGENERAL MILLS TOTAL RAISIN BR	1.6257957	0.0487126	33.375	< 2e-16
brand_feGENERAL MILLS TRIX	-0.0318812	0.0322734	-0.988	0.323231
brand_feKASHI GO LEAN CRUNCH	-3.0007301	0.0552377	-54.324	< 2e-16
brand_feKELLOGGS COCOA KRISPIES	-6.8814897	0.0940775	-73.147	< 2e-16
brand_feKELLOGGS RAISIN BRAN CRUNCH	NA	NA	NA	NA
brand_fePOST HONEYCOMB	-0.8470415	0.0409424	-20.689	< 2e-16
brand_fePOST RAISIN BRAN	2.4350469	0.0713761	34.116	< 2e-16
brand_fePOST SELECTS GREAT GRAINS	NA	NA	NA	NA
brand_feQUAKER 100% NATURAL	0.4745051	0.0297203	15.966	< 2e-16
brand_feQUAKER CAP N CRUNCH CRUNCH BE	NA	NA	NA	NA
brand_feQUAKER LIFE	NA	NA	NA	NA
brand_feGENERAL MILLS BASIC 4	NA	NA	NA	NA
brand_feGENERAL MILLS CINNAMON TST CR	NA	NA	NA	NA
brand_feGENERAL MILLS WHEATIES	NA	NA	NA	NA
(Intercept)	***			
p_jmt	***			
fiber	***			
sugars	***			
flavored	***			
fortified	***			
market_feMassachusetts	***			
market_feTexas	***			
market_feIllinois	***			
market_feIndiana	***			
manufacturer_feKELLOGGS	***			
manufacturer_fePEPSICO INC	***			
manufacturer_feRALCORP HOLDINGS	***			
time_fe5				
time_fe6	***			
time_fe7	***			
time_fe8	*			
time_fe9	***			
time_fe10	***			
time_fe11	***			
time_fe12	***			
time_fe13	***			
time_fe14	***			
time_fe15	***			

time_fe16	***
time_fe17	***
time_fe18	***
time_fe19	***
brand_feGENERAL MILLS HONEY NUT CHEER	***
brand_feGENERAL MILLS KIX	***
brand_feGENERAL MILLS LUCKY CHARMS	***
brand_feGENERAL MILLS MULTI GRAIN CHE	***
brand_feGENERAL MILLS REESES PUFFS	***
brand_feGENERAL MILLS TOTAL	***
brand_feKELLOGGS APPLE JACKS	***
brand_feKELLOGGS CORN FLAKES	***
brand_feKELLOGGS CORN POPS	***
brand_feKELLOGGS FROOT LOOPS	***
brand_feKELLOGGS FROSTED FLAKES	***
brand_feKELLOGGS FROSTED MINI WHEATS	***
brand_feKELLOGGS RAISIN BRAN	***
brand_feKELLOGGS RICE KRISPIES	***
brand_feKELLOGGS SPECIAL K	***
brand_feKELLOGGS SPECIAL K RED BERRIE	***
brand_fePOST COCOA PEBBLES	***
brand_fePOST FRUITY PEBBLES	***
brand_fePOST GRAPE NUTS	***
brand_fePOST HONEY BUNCHES OF OATS	***
brand_feQUAKER CAP N CRUNCH	***
brand_feQUAKER OATMEAL SQUARES	***
brand_feGENERAL MILLS COCOA PUFFS	***
brand_feGENERAL MILLS GOLDEN GRAHAMS	***
brand_feGENERAL MILLS TOTAL RAISIN BR	***
brand_feGENERAL MILLS TRIX	
brand_feKASHI GO LEAN CRUNCH	***
brand_feKELLOGGS COCOA KRISPIES	***
brand_feKELLOGGS RAISIN BRAN CRUNCH	
brand_fePOST HONEYCOMB	***
brand_fePOST RAISIN BRAN	***
brand_fePOST SELECTS GREAT GRAINS	
brand_feQUAKER 100% NATURAL	***
brand_feQUAKER CAP N CRUNCH CRUNCH BE	
brand_feQUAKER LIFE	
brand_feGENERAL MILLS BASIC 4	
brand_feGENERAL MILLS CINNAMON TST CR	
brand_feGENERAL MILLS WHEATIES	

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.7674 on 63907 degrees of freedom

Multiple R-squared: 0.4594, Adjusted R-squared: 0.4589

F-statistic: 936.2 on 58 and 63907 DF, p-value: < 2.2e-16

```
# Histogram of own price elasticities
reg_data <- reg_data %>%
  mutate(own_price_elasticity = coef(reg_model)["p_jmt"] * p_jmt * (1 - sj))

ggplot(reg_data, aes(x = own_price_elasticity)) +
  geom_histogram(bins = 30, fill = "steelblue", color = "black", alpha = 0.7) +
  labs(title = "Histogram of Own Price Elasticities",
       x = "Own Price Elasticity", y = "Frequency") +
  theme_minimal()
```

